BTS –VI(SS) –01.16 – 0034 Reg. No. | | | | | | | | |

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B. Tech. Degree VI Semester Special Supplementary Examination January 2016

ME 1602 MACHINE DESIGN I

(2012 Scheme)

Time: 3 Hours Maximum Marks: 100

(Use of approved Design Data Hand Book is permitted)

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) Explain the different methods to eliminate stress concentration.

(b) What is meant by endurance strength of a material? How do the size and surface condition of a component affect such strength?

(c) Show that efficiency of self locking screws is less than 50%.

(d) What is meant by preloading of bolts? Explain.

(e) Explain the phenomenon of surge in springs.

(f) Explain the various modes of failure of a riveted joint.

(g) Explain the term 'critical speed of shaft'.

(h) What are the advantages of welded joints?

PART B

(4×15 = 60)

II. A pulley is keyed to a shaft midway between two bearings. The shaft is made of cold drawn steel for which ultimate strength is 550 MPa and yield strength is 400 MPa. The bending moment at pulley varies from – 150 Nm to + 400 Nm, as the torque on shaft varies from –50 Nm to + 150 Nm. Find the diameter of the shaft. The fatigue stress concentration factors for the keyway at the pulley in bending and in torsion are 1.6 and 1.3 respectively. Take factor of safety = 1.5.

OR

III. A stepped shaft, stepped down from diameter '1.5d' to diameter 'd' with a fillet radius of 0.1d, is subjected to a torque which varies from 2000 Nm to – 800 Nm. Determine the diameter 'd'. The shaft is made of carbon steel having ultimate strength of 630 MPa and yield stress of 510 MPa.

IV. Design a cast iron unprotected type flange coupling to transmit 15 kW at 900 rpm, from an electric motor to a compressor. The maximum torque transmitted is 30% more than the mean torque. The following stresses may be taken. Shear stress for shaft, bolt and key material = 40 MPa, crushing stress for bolt and key = 80 MPa, shear stress for cast iron = 8 MPa.

OR

V. Design a socket and spigot type cotter joint to connect two rods of diameter 30 mm. The following stresses may be used. Tensile stress = 90 MPa. Shear stress = 60 MPa, crushing stress = 150 MPa.

VI. Design a valve spring of a petrol engine for the following operating conditions:

Spring load when the valve is open	=	400 N
Spring load when the valve is closed	=	250 N
Maximum inside diameter of the spring	=	25 mm
Length of the spring when the valve is open	=	40 mm
Length of the spring when the valve is closed	=	50 mm
Maximum permissible shear stress	=	400 MPa

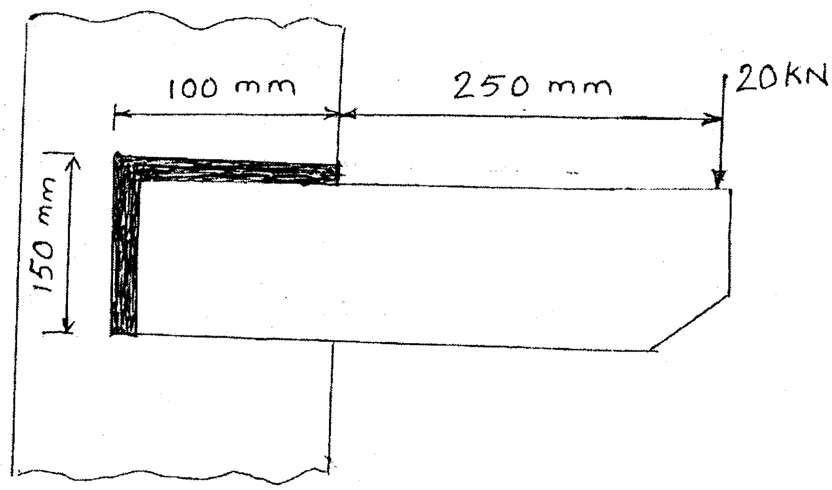
OR

VII. Design the longitudinal and circumferential joints for a boiler of 1.6 m inside diameter subjected to a pressure of 2.5 MPa. The working stresses for plates and rivets are: tensile stress = 75 MPa, shear stress = 60 MPa, crushing stress = 125 MPa. For longitudinal joint, assume triple riveted double strap butt joint with equal cover straps. The pitch of rivets in the outer row is to be double the pitch in inner row and zig zag riveting is proposed. For circumferential joint assume double riveted lap joint with zig zag riveting.

VIII. A machine shaft running at 60 rpm is supported on bearings 750 mm apart. 15 kW of power is supplied to the shaft through a 450 mm pulley located at 250 mm to the right of right bearing. The belt drive is at an angle of 60° downward with horizontal. The weight of pulley is 800 N. The angle of contact of belt is 157° and coefficient of friction is 0.4. The power is transmitted from the shaft through a 200 mm gear located 250 mm to the left of left bearing. The gear is in mesh with another gear located directly above the shaft. If the shaft material has an allowable shear stress of 67 MPa, determine the diameter of shaft. Assume pressure angle for gear as 20°.

OR

IX. Determine the size of weld for a plate welded as shown in figure. The allowable stress for the weld is 75 MPa.



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